

Assessing Professor Effectiveness (APE)

IDS CAPSTONE PROJECT REPORT

Semester: Fall 2025

Group: Pearl River Delta

Jing Yu

(Q1-2, 7, extra credit)

Elaine Zhou

(Q3-4, 8-9, title page)

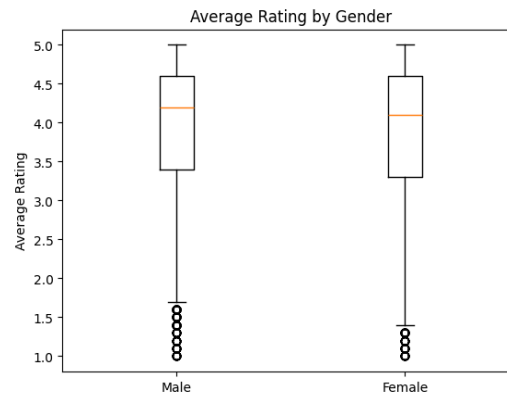
HuiJing Cong

(Q5-6, 10, collaboration setup)

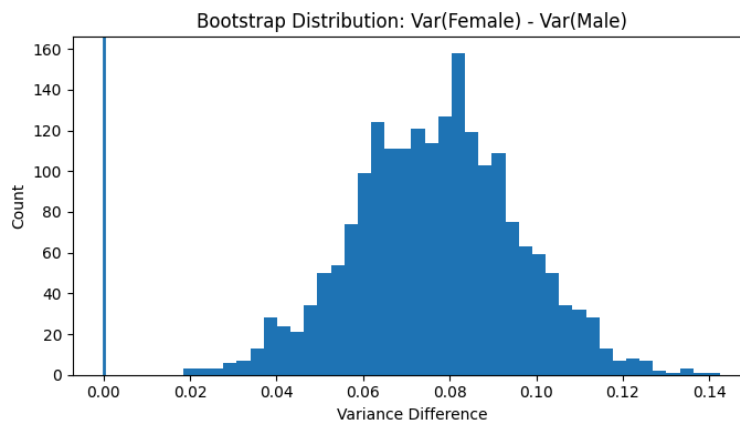
Data Cleaning and Handling:

- **Column Mapping:** Columns were manually renamed according to the project specification (mapping Num Col_1 to "AvgRating", Tags Col_1 to "ToughGrader", etc.).
- **Missing Data:** Rows containing missing values in the target column were dropped to ensure we only analyzed records with a valid target variable.
- **Filtering:** We excluded professors with fewer than five ratings and restricted gender-difference analyses to records with clearly identified gender to ensure reliability and interpretability.
- **Statistical Significance:** We used a significance threshold of 0.005 for all statistical tests.
- **RNG Seed N-number:** 11221710 (Belonging to Jing Yu)

Q1: We compared the average teaching ratings of male and female professors using a **Mann–Whitney U test**, as ratings on RMP are non-normally distributed and the two groups are independent. Professors with fewer than 5 total ratings ($\text{NumRatings} \geq 5$) were excluded to ensure stable estimates, leaving **10,015 male and 8,407 female professors**. The test indicated a statistically significant difference between genders ($U = 43,349,969$, $p = 4.90 \times 10^{-4}$). However, the effect size was very small (**Cliff's $\delta = 0.029$**). A bootstrap analysis produced a **95% confidence interval for the median difference of (0.00, 0.10)**. Overall, although the difference is statistically significant, the **effect is negligible in practical terms**, providing no strong evidence of a meaningful gender bias in average ratings.

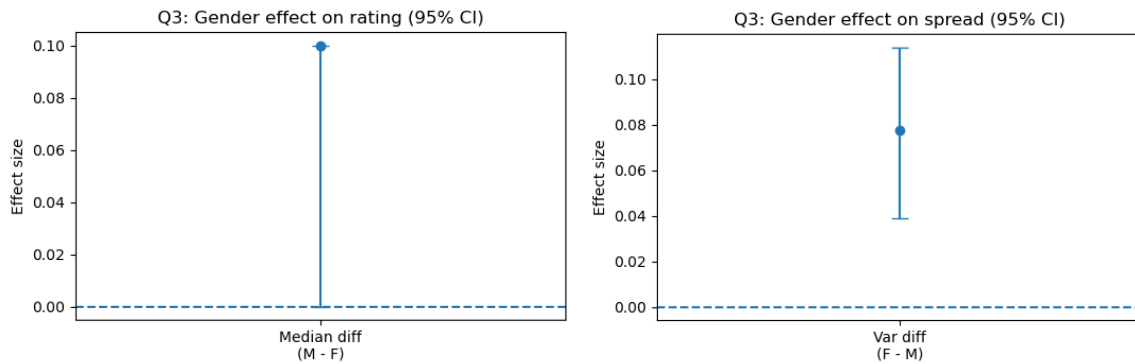


Q2: We examined differences in the dispersion of average teaching ratings between male and female professors. Professors with fewer than 5 total ratings ($\text{NumRatings} \geq 5$) were excluded to ensure stable estimates, leaving **10,015 male and 8,407 female professors**. Levene's test indicated a statistically significant difference in variance between the two groups ($p = 5.98 \times 10^{-6}$). Descriptive measures showed that female professors have slightly greater variability (**Variance: Male = 0.824, Female = 0.902; IQR: Male = 1.2, Female = 1.3**). A bootstrap analysis of the variance difference produced a **95% confidence interval for $\text{Var}(\text{Female}) - \text{Var}(\text{Male})$ of (0.039, 0.114)**. Overall, while the spread of average ratings differs statistically between genders, the **magnitude of the dispersion difference is small in practical terms**.

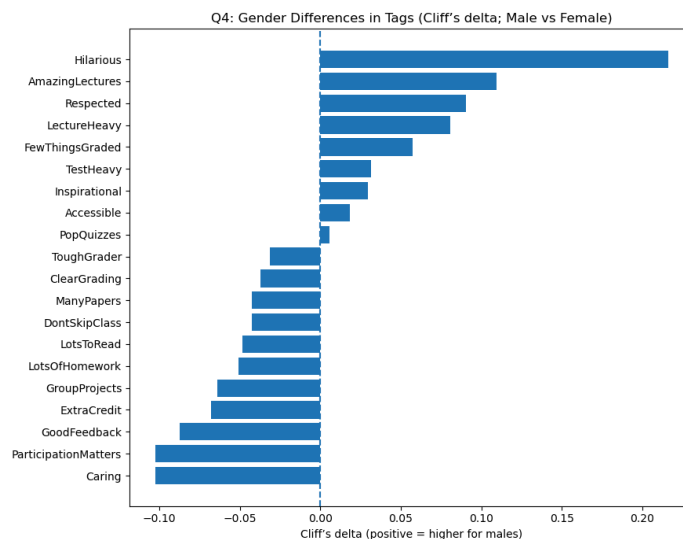


Q3: To quantify the gender effects, we estimated effect sizes and corresponding 95% CI for both effects in median and variance using **bootstrap**. Using the same filtered sample as in Questions 1 and 2 (**10,015 male and 8,407 female professors**), we measured the effect on central tendency as the **difference in medians** (male – female). The estimated median difference was **0.10**, with a **95% confidence interval of [0.00, 0.10]**, indicating an absence of evidence. The estimated variance difference was **0.0776**, with a **95%**

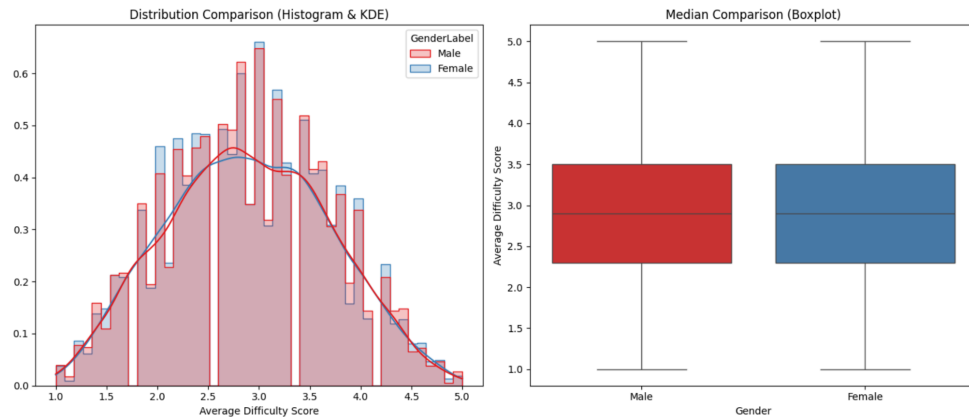
confidence interval of [0.0390, 0.1140], suggesting that ratings for female professors exhibit greater dispersion than those for male professors. By comparison, the CI for the variance difference **does not include zero**, suggesting a **more robust gender difference in rating dispersion** (female professors exhibiting greater variability in ratings). Overall, while gender differences in both central tendency and variability are detectable, the estimated effect sizes are modest.



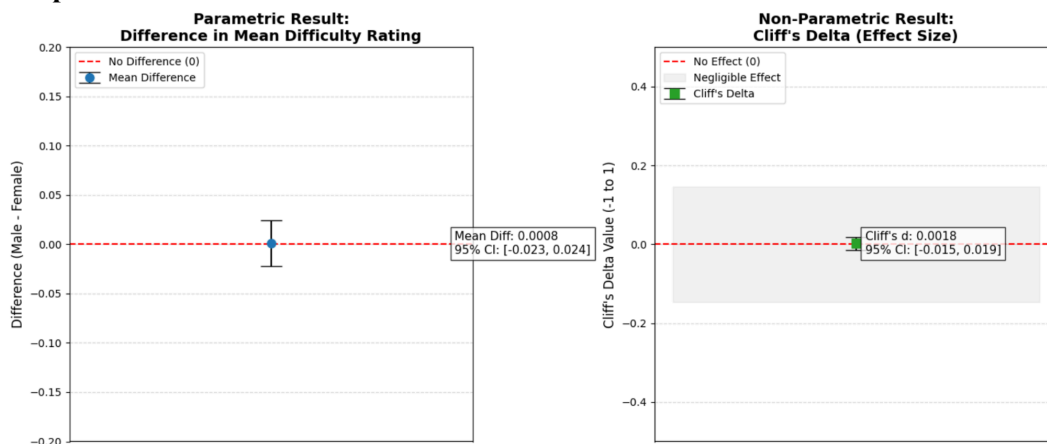
Q4: We examined whether there are gender differences in the tags awarded by students. For each of the 20 tags, we compared normalized tag proportions (tag count divided by number of ratings) between female and male professors using a **Mann–Whitney U test**. After computations, **18 out of 20 tags** exhibited statistically significant gender differences (except for **Accessible** and **Pop Quizzes**). The most gendered tags (lowest adjusted p-values) were **Hilarious** ($p = 8.32e-166$), **Amazing Lectures** ($p = 1.426e-42$), and **Participation Matters/Caring** ($p = 1.32e-34$). And the least gendered tags (highest adjusted p-values) were **Pop Quizzes** ($p = 0.308284$), **Accessible** ($p = 0.016126$), and **Inspirational** ($p = 0.000190$). These results indicate that **Hilarious** and **Amazing Lectures** were awarded more frequently to male professors, while **Participation Matters** and **Caring** were awarded more frequently to female professors. This pattern is consistent with **gendered expectations**. In contrast, the **three least gendered tags** were **Pop Quizzes**, **Accessible**, and **Inspirational**, all of which showed weak or no statistical evidence of gender differences. While many tags exhibit statistically significant gender effects, **only a subset shows practically meaningful differences**.



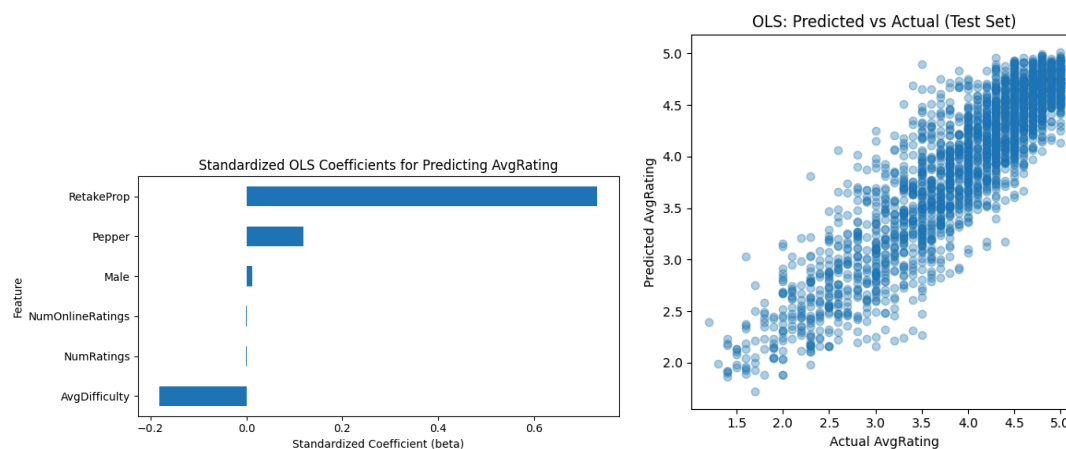
Q5: We examined whether male and female professors differ in their average difficulty ratings using non-parametric tests, as the ratings on RMP are bounded and non-normally distributed. We excluded professors with fewer than 5 total ratings ($\text{NumRatings} \geq 5$) to ensure stability in the average ratings. After filtering, our analysis included 10,015 male and 8,407 female professors. A **Mann–Whitney U test** showed no significant difference between the two groups ($p = 0.82875$), and a **Kolmogorov–Smirnov test** similarly indicated no difference in the overall distributional shapes ($p = 0.96989$). Both groups had the **same median difficulty score (2.90)**, and the histogram and boxplot confirmed substantial visual overlap. Overall, the results provide **no evidence of a gender difference** in perceived teaching difficulty.



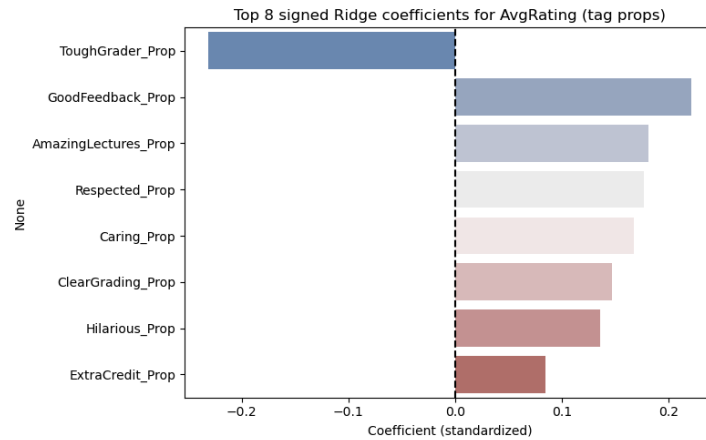
Q6: We quantified the likely size of the gender effect using both a parametric and a non-parametric effect size. With **Cohen's $d = 0.0011$** (95% CI = -0.0225 to 0.0242) and **Cliff's $\delta = 0.0018$** (95% CI = -0.0149 to 0.0189), the estimated effects are extremely small. The confidence interval for Cliff's delta was obtained using the bootstrap method (1000 resamples). Both sets of confidence intervals include zero and are tightly clustered around it. These estimates are based on Male $n = 10,015$ and Female $n = 8,407$ (same as Q5). In plain terms, the true effect is extremely small (near zero) and, with 95% confidence, lies within roughly ± 0.02 standard-deviation units (Cohen's d) or within ± 0.02 dominance probability (Cliff's delta). Because the **intervals are tightly clustered around zero** despite a large sample size, these results provide **evidence of absence** rather than merely an absence of evidence. Thus, at the 95% confidence level, the likely size of the gender difference in difficulty ratings is **effectively negligible and of no practical importance**.



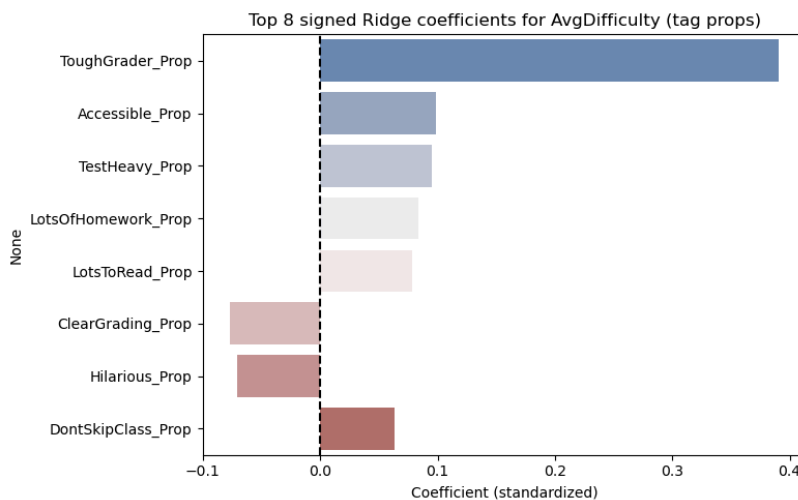
Q7: We developed a **multivariable linear regression model** to predict a professor's Average Rating using numerical features. Professors with fewer than 5 total ratings ($\text{NumRatings} \geq 5$) were excluded to ensure stable estimates, leaving **12,160 professors** in the final sample. Prior to model fitting, we assessed multicollinearity using correlation analysis and variance inflation factors (VIF), and no severe multicollinearity was detected (all $\text{VIF} < 2$). Gender was encoded using a single binary indicator (Male), with female professors as the reference group. The model achieved strong predictive performance ($R^2 = \mathbf{0.816}$, $\text{RMSE} = \mathbf{0.353}$). Standardized coefficients indicate that **RetakeProp** is the strongest predictor ($\beta \approx \mathbf{0.73}$), followed by **AvgDifficulty** ($\beta \approx \mathbf{-0.18}$) and **Pepper** ($\beta \approx \mathbf{0.12}$). The estimated gender effect was positive but small ($\beta \approx \mathbf{0.01}$), while rating volume measures contributed minimally. Overall, the results suggest that **student perceptions of course quality and difficulty are the primary drivers of teaching ratings**.



Q8: We built a **Ridge regression** model to predict a professor's Average Rating using 20 teaching-style tags. Each tag was normalized by the total number of ratings received by a professor to ensure comparability across professors. To assess potential multicollinearity, we computed **Variance Inflation Factors** (VIFs) for all tag proportions and applied Ridge regression to reduce variance from multicollinearity. The model achieved: $R^2 = \mathbf{0.7134}$ $\text{RMSE} = \mathbf{0.5009}$ in predicting average ratings. The top predictors were **Tough Grader** ($\beta = -0.231$), **Good Feedback** ($\beta = 0.222$), **Amazing Lectures** ($\beta = 0.181$), **Respected** ($\beta = 0.177$), and **Caring** ($\beta = 0.168$). Overall, the results suggest that instructional rigor tends to be penalized in student evaluations, while teaching quality and approachability are rewarded. However, compared with the numerical-predictor model in **Question 7** ($R^2 = \mathbf{0.816}$, $\text{RMSE} = \mathbf{0.353}$), it indicates that **those numerical attributes provide stronger predictive power and better explain the variability** than the tag information.

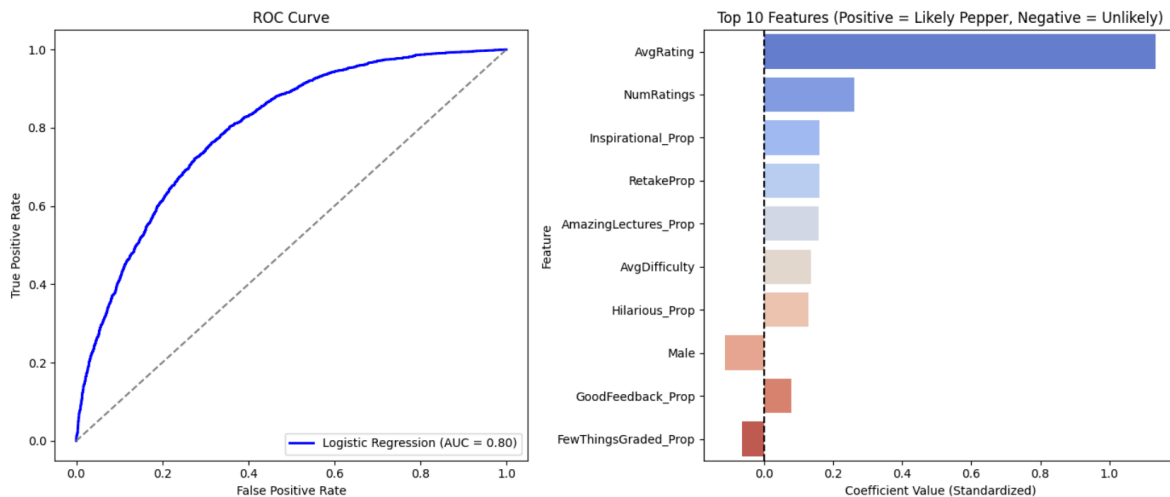


Q9: We built a **Ridge regression** model to predict a professor's Average Difficulty using the same 20 teaching-style tags. Each tag was normalized by the total number of ratings received by a professor to ensure comparability across professors. To assess potential multicollinearity, we computed **Variance Inflation Factors (VIFs)** for all tag proportions and applied Ridge regression to stabilize coefficient estimates. The model achieved $R^2 = 0.5472$ and $RMSE = 0.5421$ in predicting average difficulty. The strongest predictor was **Tough Grader** ($\beta = 0.391$), followed by **Accessible** ($\beta = 0.098$), **Test Heavy** ($\beta = 0.095$), **Lots of Homework** ($\beta = 0.083$), and **Lots to Read** ($\beta = 0.078$). In contrast, **Clear Grading** ($\beta = -0.077$) and **Hilarious** ($\beta = -0.071$) were associated with lower perceived difficulty. Overall, the results indicate that workload intensity and assessment rigor strongly increase perceived course difficulty, while clarity and an engaging classroom environment mitigate it.



Q10: We predicted whether a professor receives a “pepper” using a **logistic regression classifier** trained on all numerical variables and **normalized tag proportions**. The dataset was restricted to professors with at least five ratings and cleaned to remove missing values, and all tag counts were converted to proportions to account for differences in the number of ratings. Because pepper status is imbalanced, we used **stratified train–test splitting** and applied *class_weight='balanced'* to improve sensitivity to the minority class. After that, the model achieved strong performance with an **ROC-AUC of 0.7956** and an

overall accuracy of 0.72, correctly identifying pepper professors based on the available tags and numerical factors. The model prioritizes sensitivity, achieving a **Recall of 0.76** by correctly identifying 2,442 of 3,198 'hot' professors. This aggressive approach accepts 1,405 False Positives (**Precision 0.63**) to minimize missed detections (only 756 False Negatives), ensuring the majority of popular professors are captured. The most influential predictors were *AvgRating*, *NumRatings*, and several positive tag proportions (e.g., *Inspirational_Prop*, *RetakeProp*), while features such as *Male* and *FewThingsGraded_Prop* were negatively associated with pepper status. Overall, the model provides a reasonably accurate and interpretable characterization of the factors most strongly linked to receiving a pepper.



Q11(Extra Question): Does gender provide additional predictive value beyond other numerical features? We evaluated whether gender contributes independent predictive value for Average Rating once key numerical predictors are included. Professors with fewer than 5 total ratings ($\text{NumRatings} \geq 5$) were excluded for stability. We first fit a gender-only regression model, which explained very little variation ($R^2 = 0.010$, $\text{RMSE} = 0.931$). We then fit a full multivariable regression model including course-related predictors and gender, which achieved strong predictive performance ($R^2 \approx 0.81$, $\text{RMSE} \approx 0.35$). After adjustment, the estimated gender effect was substantially reduced ($\beta \approx 0.01$). Overall, this analysis indicates that **gender provides minimal additional predictive value beyond other numerical features**, and that differences in ratings are primarily associated with course-related characteristics rather than gender.

